Ankit Sinha

Backend Developer | TypeScript, Node.js, PostgreSQL | Open Source
sinhaankit12jan@gmail.com · +91-9971745319 · New Delhi, India · GitHub · LinkedIn

Education

Maharaja Agrasen Institute of Technology — New Delhi
B.Tech in Information Technology — CGPA: 7.69/10.0 — Sept 2023 – June 2027

Technical Skills

Languages: TypeScript, JavaScript, SQL
Backend: Node.js, Express.js, REST API Design, Zod, JWT Authentication, AuthN/AuthZ
Data & Infrastructure: PostgreSQL, Redis, Drizzle ORM, Supabase Storage, Docker Compose, Render
Async & Messaging: BullMQ (Message Queues), Background Workers, Scheduled Jobs
Reliability & Security: Transactions, Row Locks, Idempotency, Retries, Argon2, AES-256-GCM, Rate Limits
Observability: Structured JSON Logging, Request IDs, Health Checks, Graceful Shutdown
Testing & Tooling: Vitest, node:test, Integration Testing, CI/CD (GitHub Actions), Docker, Git

Open-Source Contributions

Faker.js — Merged PR #4020, v11.0 Milestone — TypeScript, Vitest, CI/CD

- Decoupled `romanNumeral` parameterized tests from internal `faker.number.int` spies using public `{min, max}` inputs.
- Improved regression-test maintainability while preserving 98.98% project coverage across CI.

FriendChise — Merged PR #398 — TypeScript, Next.js, REST API, Zod, Testing

- Built authenticated organization creation with Zod validation, demo limits, transactions, cache revalidation, and HTTP error handling.
- Added route tests for success, auth, malformed JSON, validation, usage limits, and server errors; updated API docs.

Projects

GateKeep — Encrypted Content-Sharing App — GitHub — Live

- TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, Supabase Storage, Docker Compose, Render
- Built and deployed an encrypted notes app with JWT sessions, private uploads, and a browser UI.
- Created expiring, passphrase-protected links with atomic read limits and burn-on-read behavior.
- Secured note content with AES-256-GCM and Supabase files with short-lived signed URLs.
- Added analytics, HMAC webhooks with retries, rate limits, and 28 unit/E2E tests plus integration tests.

IdleForge — Idle-Game Economy Backend — GitHub

- TypeScript, Node.js, Express.js, PostgreSQL, Drizzle ORM, Redis, BullMQ, Docker Compose
- Built atomic offline collection and transactional upgrades with guarded SQL and row locks to prevent double credit or spend.
- Added persisted idempotency for client retries and duplicate-delivery protection for BullMQ scheduled events.
- Built a Redis sorted-set leaderboard reconciled from PostgreSQL, with defined behavior during Redis outages.
- Designed separate API and workers; added correlation logs, health checks, and graceful shutdown.
- Load-tested authenticated reads at 10 concurrent (p95 114 ms on /player/me, 33 ms on /leaderboard, 0 errors, ~246 req/s mixed); CI runs typecheck, build, Docker build, and Postgres/Redis services.

Achievements & Recognition

Winner — GenAI Track, Code Kshetra 2.0 — Led backend development for the winning application.
Pre-Finalist — BVP HEX Hackathon and HackWithMAIT 6.0.